



FINAL EXAM ANALYSIS

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CIIC 5015: Artificial Intelligence

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TASK 1: OUTPUT

```
> python models.py          # Abbreviated for security.
Running shape test for AttentionBlock...
  Input  shape: torch.Size([2, 8, 16])
  Output shape: torch.Size([2, 8, 16])
Shape test passed!
```

```
Running causal mask verification...
  Causal mask verified: early positions unaffected by future token changes.

All tests passed!
```

```
> python gpt_model.py       # Abbreviated for security.
Running shape tests...

  Testing Transformer_Block...
  Transformer_Block: torch.Size([4, 16, 32]) -> torch.Size([4, 16, 32])  ✓

  Testing GPT...
  GPT forward: torch.Size([4, 16]) -> torch.Size([4, 16, 50])  ✓

  Testing generate()...
  generate: torch.Size([1, 1]) -> torch.Size([1, 11])  ✓

All shape tests passed!
```

TASK 2: CONTROL GROUP

All default values.

OUTPUT

```
Loading corpus: input.txt
Total characters : 1,115,394
Vocabulary size  : 65 unique characters
Characters       : "\n !$&',-.3:;?ABCDEFGHIJKLMNOPQRSTUVWXYZa"...
Training tokens  : 1,003,854
Validation tokens: 111,540
```

```
Model parameters : 41,921
Training for      : 3000 steps
```

Step	Train Loss	Val Loss
0	4.1743	4.1739
300	2.5734	2.5749
600	2.4633	2.4766
900	2.4098	2.4364
1200	2.3892	2.4061
1500	2.3674	2.3890
1800	2.3587	2.3804
2100	2.3484	2.3687
2400	2.3325	2.3463
2700	2.3187	2.3447
2999	2.3162	2.3343

```
=====
GENERATED SAMPLE (500 characters):
=====
```

```
Fre: mud; to so Ditoor's psind s, weriougerd s ay'st he ad lif nery trelinces Keas?
My mache hefo chirgare.
```

ARICETHESIO:

```
Girge MI she nsovesis y wayou wann:
Dit tee 'lad winciet t iss be whead peerge whe,
e co, werofpl fo, he ifopr
Toreno had we no?
Benever tsire mas merd cavath wor merunoo; teed t e, wan wo cendotoule t hano gin.
```

I mior ife ntean sene st then thandain mo ngan.

```
Whecesot Shy alare I wexpim, che bryestheld baue Re! odid st E thinghiste t
Juchil illans wicher mem;
Shy: howen s
```

```
=====
```

Training complete.

Include the loss table above and the generated sample in your analysis.pdf.

TASK 2: TREATMENT GROUP

Changed MAX_ITEERS from 3000 to 30000.

OUTPUT

Loading corpus: input.txt
Total characters : 1,115,394
Vocabulary size : 65 unique characters
Characters : "\n !\$&',-.3:;?ABCDEFGHIJKLMNOPQRSTUVWXYZa"...
Training tokens : 1,003,854
Validation tokens: 111,540

Model parameters : 41,921
Training for : 30000 steps

Step	Train Loss	Val Loss
0	4.1743	4.1739
300	2.5734	2.5749
600	2.4633	2.4766
900	2.4098	2.4364
1200	2.3892	2.4061
1500	2.3674	2.3890
1800	2.3587	2.3804
2100	2.3484	2.3687
2400	2.3325	2.3463
2700	2.3187	2.3447
3000	2.3170	2.3349
3300	2.2991	2.3380
3600	2.2997	2.3302
3900	2.2849	2.3246
4200	2.2797	2.3136
4500	2.2696	2.3032
4800	2.2639	2.2870
5100	2.2498	2.2983
5400	2.2507	2.2890
5700	2.2388	2.2813
6000	2.2448	2.2856
6300	2.2356	2.2782
6600	2.2315	2.2774
6900	2.2218	2.2695
7200	2.2201	2.2723
7500	2.2098	2.2608
7800	2.2078	2.2714
8100	2.2109	2.2606
8400	2.2019	2.2496
8700	2.1933	2.2554
9000	2.1883	2.2491
9300	2.1858	2.2521
9600	2.1912	2.2442
9900	2.1858	2.2558
10200	2.1809	2.2466
10500	2.1741	2.2465
10800	2.1697	2.2330
11100	2.1700	2.2394
11400	2.1612	2.2313
11700	2.1618	2.2281
12000	2.1572	2.2346

12300	2.1484	2.2285
12600	2.1515	2.2218
12900	2.1470	2.2246
13200	2.1426	2.2199
13500	2.1464	2.2199
13800	2.1507	2.2108
14100	2.1237	2.2101
14400	2.1291	2.2055
14700	2.1269	2.2107
15000	2.1251	2.2135
15300	2.1163	2.2185
15600	2.1244	2.2112
15900	2.1121	2.2063
16200	2.1056	2.1918
16500	2.1102	2.1937
16800	2.1047	2.2048
17100	2.1034	2.1917
17400	2.1091	2.1955
17700	2.1009	2.1901
18000	2.0957	2.1818
18300	2.1082	2.1865
18600	2.0838	2.1998
18900	2.0890	2.1965
19200	2.0835	2.1897
19500	2.0821	2.1802
19800	2.0770	2.1825
20100	2.0932	2.1877
20400	2.0844	2.1779
20700	2.0788	2.1817
21000	2.0614	2.1724
21300	2.0758	2.1625
21600	2.0708	2.1746
21900	2.0625	2.1575
22200	2.0775	2.1616
22500	2.0703	2.1715
22800	2.0630	2.1659
23100	2.0496	2.1734
23400	2.0540	2.1636
23700	2.0620	2.1613
24000	2.0610	2.1728
24300	2.0468	2.1674
24600	2.0563	2.1633
24900	2.0434	2.1624
25200	2.0478	2.1529
25500	2.0464	2.1520
25800	2.0418	2.1599
26100	2.0508	2.1529
26400	2.0332	2.1624
26700	2.0440	2.1544
27000	2.0333	2.1549
27300	2.0462	2.1492
27600	2.0365	2.1468
27900	2.0319	2.1514
28200	2.0270	2.1374
28500	2.0246	2.1444
28800	2.0375	2.1463
29100	2.0299	2.1536
29400	2.0215	2.1440
29700	2.0172	2.1424
29999	2.0221	2.1342

=====
GENERATED SAMPLE (500 characters):
=====

Frorgth womd. Nednsw!
Clor 0:
Ifult make iner as brid weath, curth me.
Gonvy bereist's; mece, anaked bangland glioom is but Qay hofaf olleld aveff durren
Mourt s APugely, grore I sthemersse, I my alllove,, toded pacesem thaindply, Aur
t cunde, at thesthat
And blougget t teittaron, balad dlay, Necrke
Thicutcoususst bak cof yoyour Nue of fownd and porro brothowin,
Ed amikse, tour me ame oin, blocow, this 'Cloysu done,
Anthis'ss
QUEEN wapp, is heffeart purpaler foreft crenow's to winduled
Ded Edin
=====

Training complete.
Include the loss table above and the generated sample in your analysis.pdf.

TASK 2 OBSERVATIONS

All verifications of Task 1 were successful.

For Task 2, I chose the Shakespeare corpus because Shakespeare's literature style is famous enough for writers to hypothesize how would Shakespeare describe modern elements.

The Control Group uses the default settings while the Treatment Group increases the total training steps from 3,000 to 30,000. In the Treatment Group, the rate of change in the training and validation losses plateaued across these increased steps, as expected. Interestingly, the sample generated by the Treatment Group seemed to mash words together and did not focus on paragraph spacing compared to the sample generated by the Control Group.